

No MCQs, only subjective questions

Paper pattern

1. A scenario based problem, students have to make a linear system of equations. No solution required.

$$6 \text{ is } 5 \text{ more than } 1 \\ 6 - 5 = 1$$

$$A + C + S = 3600 \quad (1)$$

$$A - (C + S) = 300 \\ A - C - S = 300 \quad (2)$$

$$C = 2S$$

$$C - 2S = 0 \quad (3)$$

$$A - S = C$$

$$A - C - S = 0 \quad (4)$$

A tech startup is building a cloud-based task automation platform. The project is divided among three specialized teams: **AI/ML Team**, **Cloud Infrastructure Team**, and **Security Team**. The total number of developer hours allocated across all teams is 3600 hours. During a resource planning meeting, the following constraints and relationships between the teams' workloads were decided:

- The AI/ML Team requires 300 more hours than the combined hours of the Cloud Infrastructure and Security Teams due to the integration of complex learning models.
- The Cloud Infrastructure Team must be assigned twice the hours of the Security Team, as deployment and scalability tasks are significantly more demanding.
- The AI/ML Team's workload is also tied to system risk: it was determined that the difference between the AI/ML and Security Teams' hours must be equal to the number of hours assigned to the Cloud Infrastructure Team.

Formulate a system of three linear equations in three variables. Write an Augmented Matrix but DO NOT SOLVE THE SYSTEM.

$$\begin{pmatrix} A & C & S & R \\ 1 & -1 & -1 & 3600 \\ 0 & 1 & -2 & 300 \\ 0 & -1 & -1 & 0 \end{pmatrix}$$

Practice Problems:

Problem 1: Cloud Computing Resource Allocation

A software company is allocating three types of cloud resources:

- x : number of CPU units
- y : number of GPU units
- z : number of storage units

The following conditions apply:

- The total number of allocated units is 120.
- Each GPU unit requires twice the power of a CPU unit, and each storage unit requires half the power of a CPU unit. The total power consumption equals the equivalent of 150 CPU units.
- The number of storage units is 10 more than the number of GPU units.

Task: Form a system of linear equations representing this situation.

$$x + y + z = 120$$

$$2x - y = 0$$

$$x + 2y + \frac{1}{2}z$$

Equations formed:

- $x + y + z = 120$
- $x + 2y + 0.5z = 150$
- $z = y + 10$

Problem 2: E-commerce Order Processing

An e-commerce warehouse processes three types of orders:

- x : standard orders
- y : express orders
- z : international orders

The conditions are:

- The total number of orders processed in a day is 500.
- Each express order takes twice the time of a standard order, and each international order takes three times the time of a standard order. The total processing time is equivalent to handling 900 standard orders.
- The number of express orders is 50 less than the combined number of standard and international orders.

Task: Construct a system of linear equations based on the above conditions.

Problem 3: Data Center Network Traffic

A data center handles three types of traffic:

- x : video traffic (in TB)
- y : audio traffic (in TB)
- z : text traffic (in TB)

The conditions are:

- The total traffic handled is 300 TB.
- Video traffic consumes three times the bandwidth of audio traffic, while text traffic consumes half the bandwidth of audio traffic. The total bandwidth usage is equivalent to 500 TB of audio traffic.
- The amount of video traffic is equal to the sum of audio and text traffic.

2. Solve the given system of linear equations.

$$x + 2y + z = -1, \quad 6x + y + z = -4, \quad 2x - 3y - z = 0, \quad x - y = 1$$

$$\begin{pmatrix} 1 & 2 & 1 & -1 \\ 6 & 1 & 1 & -4 \\ 2 & -3 & -1 & 0 \\ 1 & -1 & 0 & 1 \end{pmatrix}$$

There was 1 instead of zero

$$Ax = B \\ x = A^{-1}B$$

$$\begin{array}{r|rrrr} 1 & 2 & 1 & -1 \\ \hline 6 & 1 & 1 & -4 \\ - & 6 & 12 & -6 & -6 \\ \hline 0 & -11 & -5 & 2 \end{array}$$

4. Encryption and decryption

①

①

$$R_2 - 6R_1, R_3 - 2R_1, R_4 - R_1 \quad \begin{pmatrix} 1 & 2 & 1 & -1 \\ 0 & -11 & -5 & 2 \\ 0 & -7 & -3 & 2 \\ 0 & -3 & -1 & 1 \end{pmatrix}$$

②

$$-\frac{1}{11}R_2 \quad \begin{pmatrix} 1 & 2 & 1 & -1 \\ 0 & 1 & \frac{5}{11} & \frac{2}{11} \\ 0 & -7 & -3 & 2 \\ 0 & -3 & -1 & 1 \end{pmatrix}$$

$$R_1 - 2R_2, R_3 + 7R_2, R_4 + 3R_2 \quad \begin{pmatrix} 1 & 0 & \frac{1}{11} & -\frac{15}{11} \\ 0 & 1 & \frac{5}{11} & \frac{2}{11} \\ 0 & 0 & \frac{2}{11} & \frac{36}{11} \\ 0 & 0 & -\frac{4}{11} & \frac{17}{11} \end{pmatrix}$$

$$\frac{11}{2}R_3, -\frac{11}{4}R_4 \quad \begin{pmatrix} 1 & 0 & \frac{1}{11} & -\frac{15}{11} \\ 0 & 1 & \frac{5}{11} & \frac{2}{11} \\ 0 & 0 & 1 & 18 \\ 0 & 0 & -1 & -\frac{17}{4} \end{pmatrix}$$

$$R_4 - R_3 \quad \begin{pmatrix} 1 & 0 & \frac{1}{11} & -\frac{15}{11} \\ 0 & 1 & \frac{5}{11} & \frac{2}{11} \\ 0 & 0 & 1 & 18 \\ 0 & 0 & 0 & -\frac{87}{4} \end{pmatrix}$$

$$-\frac{17}{4} - 18$$

No soluti

Let y

$$\begin{pmatrix} 1 & 0 & \frac{1}{11} & -\frac{15}{11} \\ 0 & 1 & \frac{5}{11} & \frac{2}{11} \\ 0 & 0 & 1 & 18 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$z = 18$$

$$y + \frac{5}{11}z = 2$$

$$x + \frac{1}{11}z = -\frac{15}{11}$$

$$x = -\frac{15}{11} - \frac{18}{11}$$

$$y = \frac{90}{11} - 2$$

$$y = 2 - \frac{90}{11}$$

$$\begin{array}{rrrr} -1 & - & - & + \\ 0 & -11 & -5 & 2 \end{array}$$

$$R_3: 2 \quad -3 \quad -1 \quad 0$$

$$2R_1: 2 \quad -4 \quad 2 \quad -2$$

$$\begin{array}{rrrr} 1 & -\frac{10}{11} & -1 & -\frac{4}{11} \\ 0 & -7 & -3 & 2 \end{array}$$

③

$$\begin{pmatrix} 1 & 2 & 1 & -1 \\ 0 & 2 & \frac{10}{11} & \frac{4}{11} \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & \frac{1}{11} & -\frac{15}{11} \\ 0 & -7 & -3 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 0 & -7 & -3 & 2 \\ 0 & 7 & \frac{35}{11} & \frac{14}{11} \end{pmatrix}$$

$$\begin{pmatrix} 0 & 0 & \frac{2}{11} & \frac{36}{11} \\ 0 & 0 & \frac{2}{11} & \frac{36}{11} \end{pmatrix}$$

$$R_4: 0 \quad -3 \quad -1 \quad 1$$

$$3R_2: 0 \quad 3 \quad \frac{15}{11} \quad \frac{6}{11}$$

$$\begin{pmatrix} 0 & 0 & -\frac{4}{11} & \frac{17}{11} \\ 0 & 0 & -\frac{4}{11} & \frac{17}{11} \end{pmatrix}$$

$$\frac{36}{11} \times \frac{11}{2}$$

$$\frac{17}{11} \times -\frac{11}{4}$$

$$x = -1, y = -2, z = 4$$

(x, y, z) = (-1, -2, 4)

is solution of above equations. verify

3. Euclidean transformation

The equation of ellipse is given below. Find the new equation after rotating it through an angle of 90° , and sketch both the original and the rotated ellipse.

$$\frac{(x-3)^2}{16} + \frac{(y+2)^2}{9} = 1$$

- ellipse
- parabola
- circle

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos 90^\circ & -\sin 90^\circ \\ \sin 90^\circ & \cos 90^\circ \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

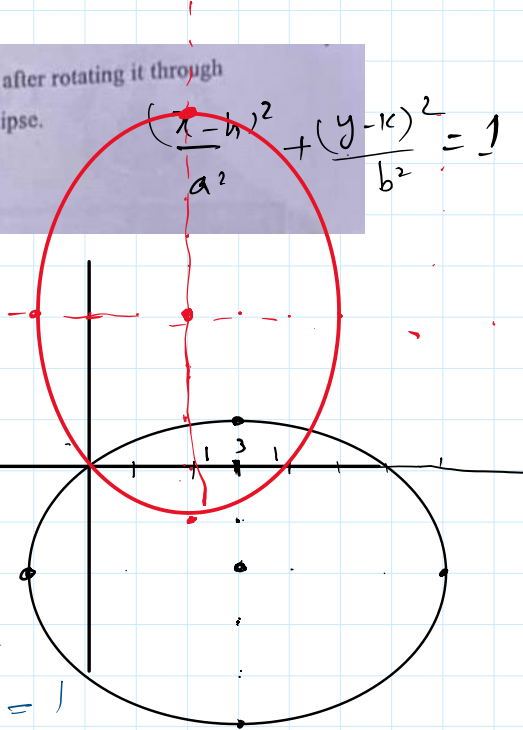
$$x' = -y$$

$$y' = x$$

$$\frac{(y'-3)^2}{16} + \frac{(-x'+2)^2}{9} = 1$$

$$\frac{(-3)^2 (x'-2)^2}{3^2} + \frac{(y'-3)^2}{4^2} = 1$$

$$\frac{(x'-2)^2}{3^2} + \frac{(y'-3)^2}{4^2} = 1$$



Question 1: Ellipse (Rotation + Sketch)

The equation of an ellipse is given by

$$\frac{(x-4)^2}{25} + \frac{(y+1)^2}{9} = 1$$

Tasks:

1. Find the new equation after rotating the ellipse through 90° about the origin.
2. Identify the new center and orientation of axes.
3. Sketch both the original and rotated ellipse.

Question 2: Parabola (Transformation + Geometry)

The equation of a parabola is

$$(y-2)^2 = 8(x-1)$$

Tasks:

1. Rotate the parabola by 90° counterclockwise about the origin and find the new equation.
2. Determine the vertex, focus, and directrix of both the original and rotated parabola.
3. Sketch both parabolas.

Question 3: Circle (Rotation + Invariance Concept)

The equation of a circle is

$$(x-2)^2 + (y-3)^2 = 16$$

Tasks:

1. Rotate the circle through 90° about the origin and find the new equation.
2. Compare the original and rotated equations. What do you observe?
3. Sketch both circles and comment on symmetry.

Question No. 4 [CLO2] (3+3+2 = 8 Marks)

You are part of a 2D game development team working on a side-scrolling adventure game. Create the initial shape of the character using the following polygons:

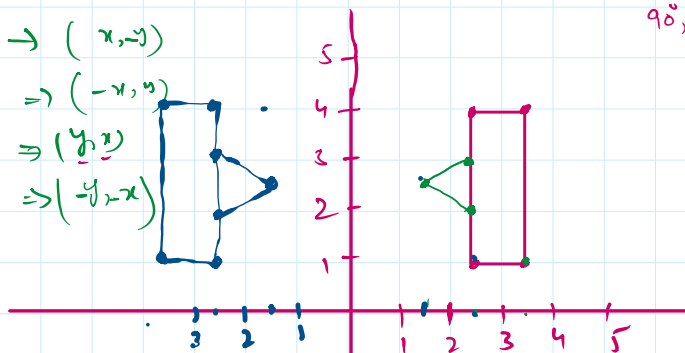
Body (Rectangle): Vertices at (2.5, 1), (2.5, 4), (3.5, 4), (3.5, 1)

Left Arm (Rectangle): Vertices at (2.5, 2), (2.5, 3), (1.5, 2.5)

Find the new coordinates after applying reflection over the y-axis, and sketch both the original and the reflected character.

Reflection

- ✓ x-axis $(x, y) \rightarrow (x, -y)$
- ✓ y-axis $(x, y) \rightarrow (-x, y)$
- ✓ $y = x$ $(x, y) \Rightarrow (y, x)$
- ✓ $y = -x$ $(x, y) \Rightarrow (-y, -x)$



Rotation

90° clock
 $\begin{pmatrix} \cos & -\sin \\ \sin & \cos \end{pmatrix}$

x-axis $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

y-axis $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$

$y = x$ $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$y = -x$ $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$

Short way Rotation of Co-ordinates

90° Anticlock $(x, y) \rightarrow (-y, x)$

$(2, -3) \rightarrow (+3, 2)$

$(-2, -5) \rightarrow (+5, -2)$

90° clockwise $(x, y) \rightarrow (y, -x)$

180° $(x, y) \rightarrow (-x, -y)$

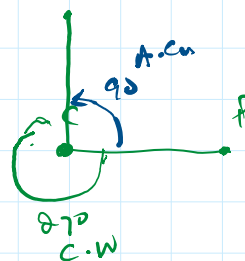
Translation

$= \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} a \\ b \end{pmatrix}$

$\begin{pmatrix} -2 \\ 3 \end{pmatrix} + \begin{pmatrix} 5 \\ 0 \end{pmatrix}$

$= \begin{pmatrix} 3 \\ 3 \end{pmatrix}$

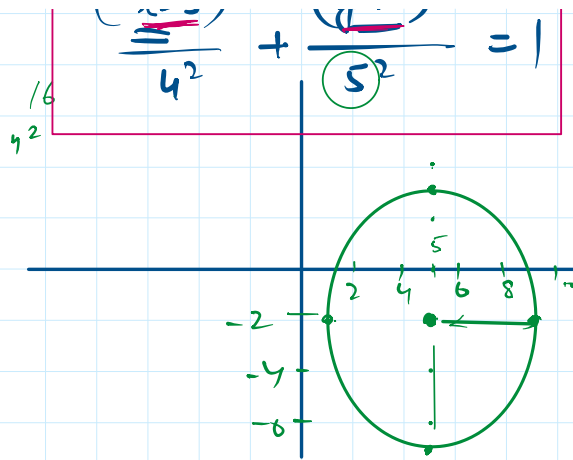
270° clockwise = 90° A.C.W
 270° A.C.W = 90° C.W



$\frac{(x-5)^2}{4^2} + \frac{(y+2)^2}{5^2} = 1$

$(5, -2)$

$x-5=0$
 $x=5$



$$\begin{aligned}
 x-5 &= 0 \\
 x &= 5 \\
 y+2 &= 0 \\
 y &= -2 \\
 (5, -2)
 \end{aligned}$$

4. Encryption and decryption

Modulo
+ve

$$\begin{aligned}
 2578 &\Rightarrow 2578 \\
 &\quad - 2574 \\
 \hline
 99 \times 26 &\Rightarrow \textcircled{4} \\
 2578 &\equiv 4 \pmod{26}
 \end{aligned}$$

$\textcircled{4} \cdot 68$
x
26

-ve

$$\begin{aligned}
 -24 + 26 &= 2 (B) \\
 -26 + 26 &= 0 + 26 = 26 (Z) \\
 -59 & \\
 3 \times 26 &= \underline{78} \\
 &\quad 19
 \end{aligned}$$

Reciprocal

$$\begin{aligned}
 a) \quad \frac{1}{3} &\equiv K \\
 \frac{1}{5} & \\
 \frac{1}{15} &
 \end{aligned}$$

$3K \quad (26)$
R=1

$4 \cdot 259$
 $\downarrow$
 $u+1 = 5 \times 26$

$$\begin{aligned}
 -359738 + 359762 \\
 \hline
 13837 \\
 26 \times
 \end{aligned}$$

$= 24$

For future reference, in Table 2 we provide the following reciprocals modulo 26:

Table 2 Reciprocals Modulo 26

a	1	3	5	7	9	11	15	17	19	21	23	25
a ⁻¹	1	9	21	15	3	19	7	23	11	5	17	25

$$\begin{aligned}
 \frac{1}{15} &= 15^{-1} = 7 \\
 &= \textcircled{-} \frac{1}{5} \begin{pmatrix} 6 & -2 \\ 3 & 8 \end{pmatrix} = \textcircled{-} \begin{pmatrix} -6 & 2 \\ -3 & -8 \end{pmatrix} = \begin{pmatrix} -126 & 42 \\ -63 & -168 \end{pmatrix} \\
 &= \begin{pmatrix} 4 & 17 \\ 15 & 14 \end{pmatrix}
 \end{aligned}$$

$$= \begin{matrix} 4 & 11 \\ 15 & 14 \end{matrix}$$

Question No. 5 [CLO2] (4+4 = 8 Marks)

As part of a secure messaging protocol in a cyber-defense training exercise, you are asked to encrypt and decrypt a short message applying the Hill Cipher method. Encode and decode the message "BUG HUNTER".

- Convert the message into numerical form using

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26=0

- Use the following key matrix for encryption

$$K = \begin{bmatrix} 3 & 3 \\ 1 & 6 \end{bmatrix} \begin{pmatrix} 2 & 7 & 21 & 20 & 18 \\ 21 & 8 & 14 & 5 & 18 \end{pmatrix}$$

Good Luck!

$\begin{pmatrix} B & G & U & T & R \\ U & H & N & E & R \end{pmatrix}$

QXSCAAWxDV

$$= \begin{matrix} 69 & 45 \end{matrix}$$

69	45	105	75	108
128	55	105	50	126

→ under mod 26

17	19	1	23	4
24	3	1	24	22

$$K P = C$$

$$P = K^{-1} C$$

$$K^{-1} = \frac{1}{15} \begin{pmatrix} 6 & -3 \\ -1 & 3 \end{pmatrix}$$

$$= 7 \begin{pmatrix} 6 & -3 \\ -1 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 42 & -21 \\ -7 & 21 \end{pmatrix} \text{ under mod } 26$$

$$= \begin{pmatrix} 16 & 5 \\ 19 & 21 \end{pmatrix}$$

17	19	1	23	4
24	3	1	24	22

392	319	21	488	174
827	424	40	941	538

2	7	21	20	18
21	8	14	5	18

BUG HUNTER

do this

MIDTERM

$$K^{-1} = \begin{pmatrix} -1 & 3 \\ 5 & 4 \end{pmatrix}$$

LINEAR ALGEBRA

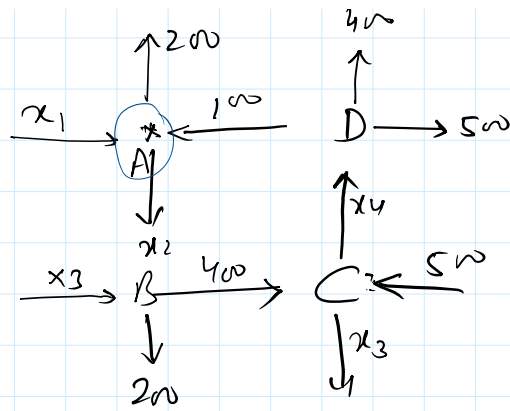
ALGEBRA

MATHEMATICS

University of Central

$$\begin{matrix} \uparrow 200 & \uparrow 400 \\ 100 & \end{matrix}$$

$$A: x_1 + 100 = 200 + x_2$$



$$A: x_1 + 100 = 200 + x_2$$

$$B: x_2 + x_3 = 200 + 400$$

$$C: 400 + 500 = x_3 + x_4$$

$$D: x_4 = 100 + 500 + 400$$

$$x_1 - x_2 = 100$$

$$x_2 + x_3 = 600$$

$$x_3 + x_4 = 900$$

$$x_4 = 1000$$

$$\begin{pmatrix} x_1 & x_2 & x_3 & x_4 & = & C \\ 1 & -1 & 0 & 0 & & 100 \\ 0 & 1 & 1 & 0 & & 600 \\ 0 & 0 & 1 & 1 & & 900 \\ 0 & 0 & 0 & 1 & & 1000 \end{pmatrix}$$